Machine Learning Theory & Practical Lab Manual

Master of Computer Application
Semester – II

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EXPERIMENT 03

SIMPLE AND MULTIPLE LINEAR REGRESSION ANALYSIS ON CONCRETE STRENGTH DATASET

1. OBJECTIVE

The purpose of this lab is to study how linear regression works in practice. The experiment focuses on implementing both **Simple Linear Regression** and **Multiple Linear Regression** in Python. The aim is to analyze how different ingredients used in concrete mixtures influence the **compressive strength of concrete** and evaluate how accurately a regression model can predict that strength.

2. DATASET DESCRIPTION

The experiment uses the Concrete Compressive Strength Dataset. This dataset contains measurements of various materials used in concrete mixtures and the resulting compressive strength.

Each record represents a particular concrete mix and includes quantities of materials such as cement, water, aggregates, and additives. The dataset also records the age of the concrete sample when the strength was measured.

Main Variables

- Cement
- Blast Furnace Slag
- Fly Ash
- Water
- Superplasticizer
- Coarse Aggregate
- Fine Aggregate
- Age (in days)
- Concrete Compressive Strength (target variable)

The dataset contains **1030 observations and 9 attributes**.

3. SOFTWARE AND LIBRARIES USED

The experiment was performed using Python along with several libraries that support data analysis and machine learning.

- **Pandas** – used for loading and manipulating the dataset
- **NumPy** – used for numerical computations
- **Matplotlib** and **Seaborn** – used to visualize the data
- **Scikit-learn** – used to build regression models and evaluate their performance

4. EXPLORATORY DATA ANALYSIS (EDA)

Before building models, the dataset was explored to understand its structure and statistical properties.

Steps Performed

1. The dataset was loaded using pandas.
2. The first few rows were displayed to inspect the data.
3. The dimensions of the dataset were checked.
4. Summary statistics such as mean, standard deviation, minimum, and maximum values were calculated.
5. Missing values were checked.

Observations

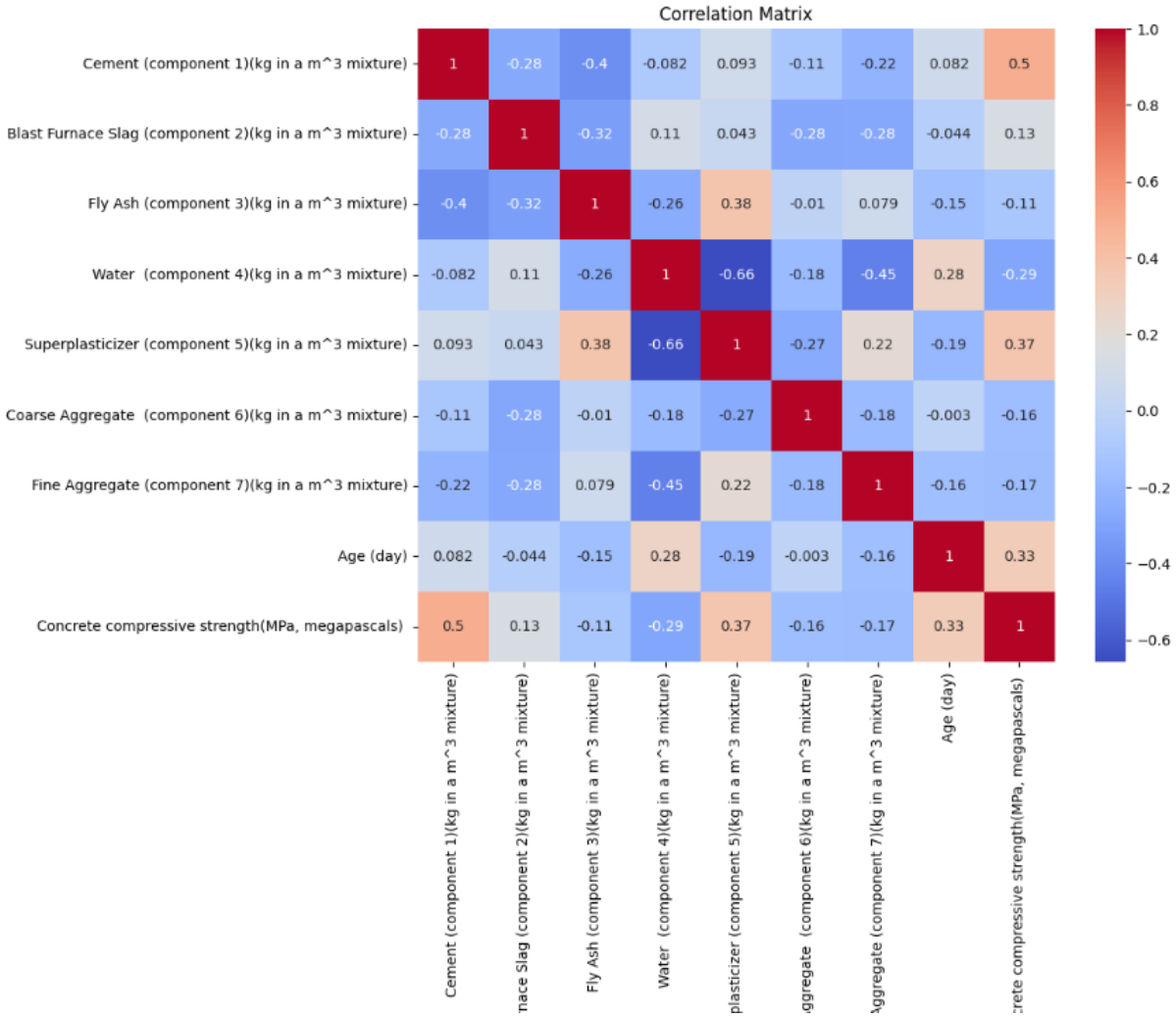
The dataset contained **no missing values**, so no imputation was required. However, some variables had a wide range of values, especially **Age and Superplasticizer**, indicating high variability in the mixtures.

Visual Analysis

Several visualizations were created to understand the relationships among variables:

- Histograms to observe data distribution
- A correlation heatmap to identify relationships between features
- Pair plots to explore interactions between variables

These visualizations helped identify which variables might influence compressive strength.



5. DATA CLEANING AND TRANSFORMATION

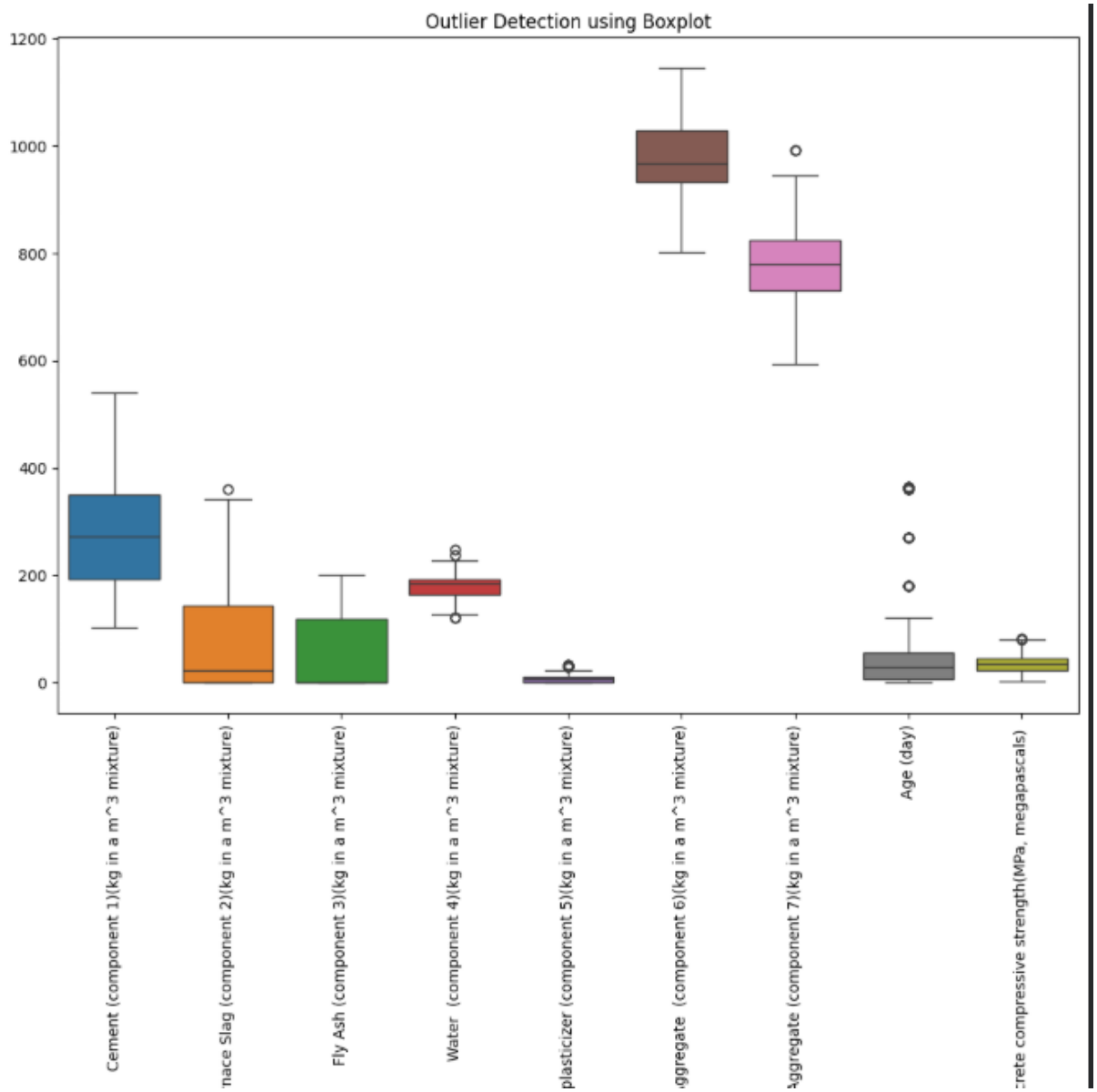
Handling Missing Values

After checking the dataset, it was found that there were **no missing values**, so the dataset could be used directly.

Outlier Detection

Outliers were identified using the **Interquartile Range (IQR) method**. Some variables, including water, superplasticizer, fine aggregate, and age, showed extreme values.

A higher number of outliers appeared in the **Age variable** because the dataset includes samples measured from **1 day up to 365 days**. These values are valid experimental observations and therefore were not removed.



Feature Scaling

To ensure that variables with larger numeric ranges did not dominate the model, feature scaling was performed using **StandardScaler**. This standardizes the data so that each feature has a mean of zero and a standard deviation of one.

6. SIMPLE LINEAR REGRESSION

In this part of the experiment, only **one independent variable** was used to predict the compressive strength. Cement was chosen as the predictor because it has a strong influence on concrete strength.

Procedure

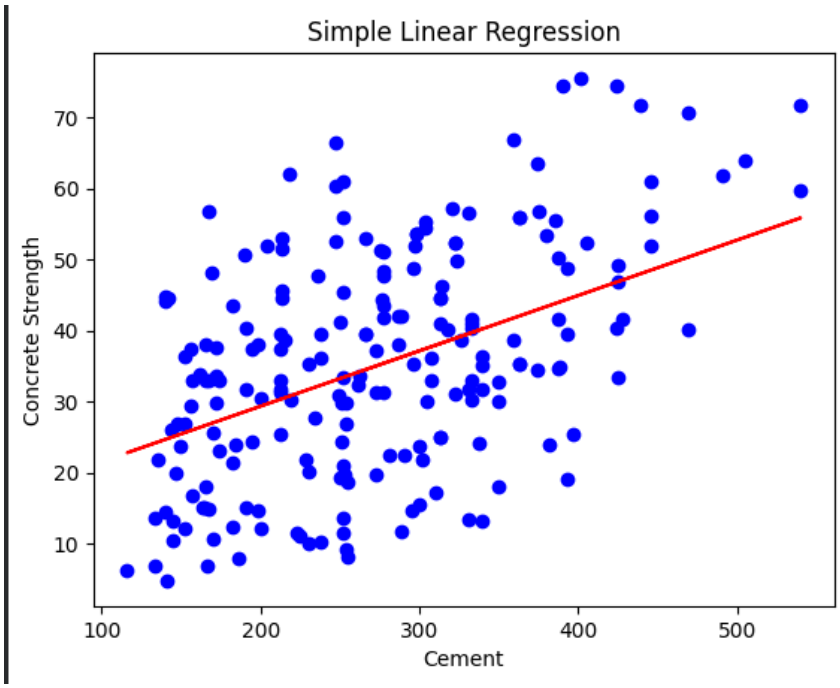
1. Cement was selected as the independent variable.
2. The dataset was divided into training and testing sets.
3. A linear regression model was trained using the training data.
4. Predictions were made using the test dataset.
5. A regression plot was generated to visualize the relationship between cement and compressive strength.

Results

Metric	Value
R2 Score	0.25
MAE	11.55
MSE	192.78
RMSE	13.88

Discussion

The R^2 value indicates that the model explains around **25% of the variability** in compressive strength. Since only one variable was used, the model cannot fully capture the factors affecting concrete strength.



7. MULTIPLE LINEAR REGRESSION

To improve prediction accuracy, a multiple regression model was created using **all available features**.

Procedure

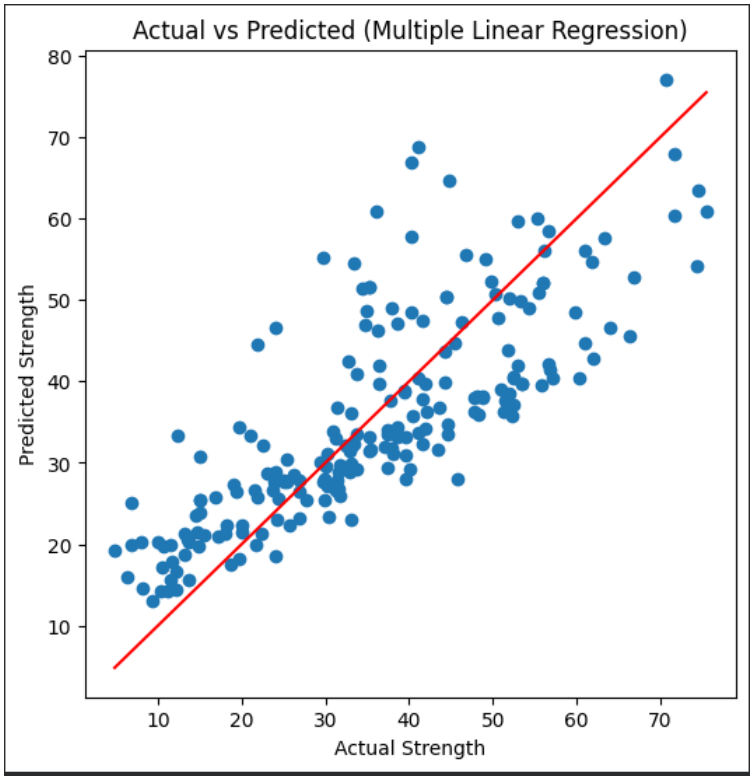
1. All ingredient variables and age were used as input features.
2. The dataset was divided into training and testing sets.
3. A multiple linear regression model was trained.
4. Predictions were generated for the test data.
5. Actual values were compared with predicted values using a scatter plot.

Results

Metric	Value
R2 Score	0.63
MAE	7.74
MSE	95.97
RMSE	9.79

Discussion

The multiple regression model produced a significantly higher R^2 score compared to the simple regression model. This indicates that the compressive strength of concrete depends on **multiple factors**, including material composition and curing time.



8. COMPARISON OF MODELS

Model	R2 Score	RMSE
Simple Linear Regression	0.25	13.88
Multiple Linear Regression	0.63	9.79

The comparison shows that the **Multiple Linear Regression model performs better**, as it takes into account several variables rather than relying on a single predictor.

9. CONCLUSION

This experiment demonstrated the application of linear regression techniques to predict concrete compressive strength. The simple regression model provided a basic understanding of the relationship between cement content and strength, but its predictive capability was limited.

When multiple variables were included, the model achieved improved accuracy. This indicates that concrete strength is influenced by a combination of ingredients and curing conditions. Therefore, multiple regression provides a more realistic representation of the underlying relationships in the dataset.